

$$a) \frac{dy}{dx} = 3a(x-1)^2 + 2b(x-1) + c$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 0, \text{ so } c = 0.$$

$$b) \begin{array}{l|l} -a + b + 2 = 1 & ab = 6 \\ -a + b = -1 & b = \frac{6}{a} \\ -a + \frac{6}{a} = -1 & \\ -a^2 + a + 6 = 0 & \\ a = -2 \text{ or } a = 3 & \\ b = 3 \text{ or } b = 2 & \end{array}$$

$$\frac{d^2y}{dx^2} = 6a(x-1) + 2b$$

$$\left. \frac{d^2y}{dx^2} \right|_{x=1} < 0, \therefore \begin{cases} a = -2 \\ b = 3 \end{cases}$$

thus the claim is disagreed.

$$\frac{dy}{dx} = -b(x-1)^2 - b(x-1)$$

$$\text{when } \frac{dy}{dx} = 0, (x-1)(-b(x-1) - b) = 0$$

$$x = 1 \text{ or } x = 2.$$